ALICE L WALTON SCHOOL OF MEDICINE

Management of Shock

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Adapted from: Management of Shock

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the hemodynamic findings for each type of shock.
2. Explain the use of fluid resuscitation in shock.
3. Justify the use of pharmacological agents in shock treatment
4. Outline the treatment plan for each type of shock

CASE CONNECTION

LT is brought to the emergency department by his son. “My dad lives with me. For the past few days, he has been confused, tired, and coughing up green stuff. This morning he said he wanted to get ready for his wedding, and Mom died 15 years ago.” On exam, LT is lethargic, and his vitals are a blood pressure of 80/62 mm Hg, heart rate is 116 beats/min, respiratory rate is 30 breaths/min, and temperature is 101.3°F. On physical exam, he has crackles at the right lower lung base, and his skin is cool. You start IV fluids, and labs and imaging are ordered. LT’s white blood cell count and serum creatinine are markedly elevated. His chest x-ray demonstrates right lower lobe pneumonia. After receiving 2 L of IV fluids in the emergency department, LT becomes more lethargic, and his blood pressure does not improve.

What is causing LT’s hemodynamic condition? What medications should be given next? Consider your answers as you read, and we’ll revisit LT at the end of the brick.

GO TO CONCLUSION ↓

What Is Shock? (LO#1)

Shock occurs when blood flow to the body’s tissues decreases below their oxygenation needs, which is most easily quantified by hypotension.

Clinical Significance

Shock is defined as a sues, resulting in widespread cellular and tissue hypoxia. This shortage is not immediately fatal, but it can rapidly progress to multiorgan system failure and death if not corrected quickly. This brick will focus on how we stabilize the patient while looking for the underlying cause of shock (eg, infection, trauma, or heart failure).

Types of Shock

The five types of shock are:

- **Distributive shock** arises from systemic vasodilation, which might be triggered by conditions such as sepsis or anaphylaxis.
- **Cardiogenic shock** can occur from a condition such as heart failure or myocardial infarction that leads to low stroke volume (SV) and low cardiac output (CO).
- **Hypovolemic shock** is caused by low intravascular volume that leads to low preload and low CO and can be caused by hemorrhage, GI blood loss, or severe dehydration.
- **Neurogenic shock** is caused by the loss of sympathetic output from the central nervous system (CNS), leading to a global loss of arterial tone, cardiac output, heart rate, and preload (venous tone); neurogenic shock can follow head or spinal cord trauma.
- **Obstructive shock** is due to an obstruction in blood flow through the heart. There are two types. In the pulmonary type, a pulmonary embolus causes increased resistance to flow within the pulmonary circulation and prevents blood from reaching the left side of the heart. In the structural type, there is global obstruction of flow through the heart caused by cardiac tamponade or tension pneumothorax.

type of shock is shown with an asterisk.

In the previous session, we discussed preload, afterload, cardiac output (CO), and mean arterial pressure (MAP) as the primary physiologic determinants of perfusion. In clinical practice, particularly in critically ill patients, additional hemodynamic measurements help us better classify shock and guide therapy. These include **right atrial pressure (RAP), pulmonary capillary wedge pressure (PCWP), cardiac index (CI), and mixed venous oxygen saturation (SvO_2).** Together, these measurements help localize where in the circulatory system the problem is occurring — venous return, ventricular filling, pump function, or tissue oxygen utilization.

Table 1 Hemodynamic findings of shock

	Mean arterial blood pressure (MAP)	End-diastolic volume (preload)	Systemic vascular resistance	Cardiac output	Heart rate
Distributive	↓	↓	↓P	↑	↑
Cardiogenic	↓	↑	↑	↓P	↑
Hypovolemic	↓	↓P	↑	↓	↑
Neurogenic	↓P	↓P	↓P	↓P	*Normal or ↓
Obstructive: pulmonary	↓	*↑ (RV) ↓ (LV)	↑	↓P	↑
Obstructive: mechanical	↓	↓	↑	↓P	↑

Right Atrial Pressure Right atrial pressure reflects the pressure in the

CLINICAL REASONING

PCWP, like RAP helps differentiate cardiogenic and obstructive shock from hypovolemic shock.

Cardiac Index (CI) Cardiac index adjusts cardiac output for body surface area

CI = CO/BSA

Normal CI is approximately **2.5–4.0 L/min/m²**. A reduced CI indicates inadequate forward flow relative to patient size and is seen in cardiogenic, hypovolemic, and obstructive shock. In early distributive (septic) shock, CI may be normal or elevated due to decreased systemic vascular resistance and compensatory hyperdynamic circulation. (hyperdynamic = decrease in SVR due to endotoxin release leads to increase in CO, however, in late septic shock SVR may increase)

Mixed Venous Oxygen Saturation (SvO_2) Mixed venous oxygen saturation reflects the percentage of oxygen remaining in blood returning to the heart after tissue extraction. It represents the balance between oxygen delivery (DO_2) and oxygen consumption (VO_2). Normal SvO_2 ranges from **60–80%.** Low SvO_2 indicates that tissues are extracting more oxygen because delivery is inadequate (low cardiac output states such as cardiogenic or hypovolemic shock). Elevated SvO_2 may be seen in early septic shock due to impaired oxygen extraction at the tissue level.

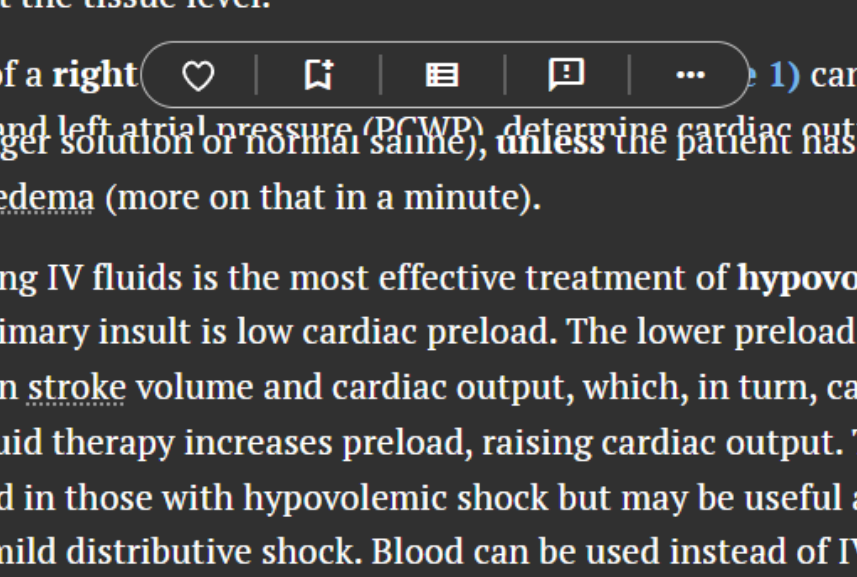
Placement of a **right** 1) can determine right atrial and left atrial pressure (PCWP), determine cardiac output, and allow

lactated Ringer solution or normal saline), **unless** the patient has evidence of pulmonary edema (more on that in a minute).

Administering IV fluids is the most effective treatment of **hypovolemic shock.** Here, the primary insult is low cardiac preload. The lower preload causes reductions in stroke volume and cardiac output, which, in turn, cause low blood pressure. Fluid therapy increases preload, raising cardiac output. The effect is most marked in those with hypovolemic shock but may be useful as well in those with mild distributive shock. Blood can be used instead of IV fluids but is generally reserved for those with overt hemorrhage.

Giving IV fluids carries little risk, which is why IV fluids are used in most hypotensive patients, even before the precise cause of low blood pressure or shock is identified. In fact, the patient’s response to saline is often diagnostic: if they respond promptly, you are probably dealing with hypovolemic shock and not one of the other types.

When should you **not** give IV fluids? Patients should not receive IV fluids if they have pulmonary edema, diagnosed hypoxia, and interstitial edema on chest x-ray (Figure 1).



How Do We Manage Distributive Shock with Vasopressors? (LO#3)

As you see in Table 1, the different forms of shock have different physiologic patterns. Many of the therapies for shock are directed specifically against one of these abnormalities. We will now work through some of these, beginning with reduced systemic vascular resistance (SVR), one of the characteristics of distributive shock.

Distributive shock is caused by arteriolar vasodilation, which reduces SVR. Therefore, **vasopressor drugs** (sometimes called “pressors”), which cause constriction of the arterioles, are an effective treatment. The following is a list of receptors and the pressor drugs that activate them to achieve vasoconstriction:

- α_1-Adrenergic, activated by norepinephrine, epinephrine, or phenylephrine
- Vasopressin V_{1A}, activated by vasopressin
- Dopaminergic, activated by dopamine

Norepinephrine

Although the use of these pressor drugs varies and can be customized for an individual patient, norepinephrine is the drug of choice for most patients with distributive shock, as has been shown in survival studies. It acts primarily at the α_1 adrenergic receptor, secondarily at the β_1, and β_2 receptors. The drug causes vasoconstriction and tachycardia (via β_1 stimulation).

An adverse effect is tachyarrhythmias, although the risk is lower than that of stronger β_1 agonists. Like the other pressor drugs, norepinephrine should only be used when absolu the drugs may

is to induce vasodilation in the renal circulation. The degree of activity on each type of receptor is dose-dependent: low doses are primarily dopaminergic, and increasing doses produce inotropic effects via β_1 stimulation, then systemic vasoconstriction α_1 stimulation. Side effects include tachycardia and tachyarrhythmias, as is the case with most drugs with inotropic effects.

CLINICAL CORRELATION

Because acute kidney injury is so common in shock-states, the theoretical benefit of low-dose dopamine is to preserve blood flow to the kidneys that sustains renal function. However, there has been no proven renal-protective benefit in the clinical setting, and increasing renal blood flow may lead to diuretic effect that can cause significant harm in a hypotensive patient.

These drugs can be used in combination, and it is common to mix drugs with different receptor targets (eg, use norepinephrine with vasopressin). Table 2 summarizes the effects of the vasopressor drugs on the various receptors; the number of plus signs indicates relative receptor affinity. (**This is a must know table !!)**

Table 2 Vasopressor receptor targets

Drug	α_1	β_1	β_2	V_{1A}	Dopaminergic
Norepinephrine	[illegible]	[illegible]	[illegible]	[illegible]	None

preferred)

- Bolus strategy: Rapid IV boluses (500-1000mL), reassess, repeat as needed
- Blood products if bleeding: If hemorrhagic shock, transition to packed red blood cells (pRBCs) early.
- Monitor hemodynamic and organ perfusion response

Treatment: Distributive Shock

- Septic Shock (most common):- Fluids first: 30mL/kg crystalloid bolus within 3 hours (well-established initial resuscitation)
- Antibiotics: Administered as soon as possible
- Vasopressors: Norepinephrine is first-line when fluids aren't enough
- Source control: Find and fix the infection (drain abscess, remove infected catheter, etc.)

Treatment: Anaphylactic Shock

- Epinephrine: first-line (EpiPen) to reverse vasodilation (α-1 mediated vasoconstriction will reverse) and bronchospasm (β2 mediated bronchodilation will reverse)
- Fluids: Aggressive IV crystalloids
- Adjuncts: H1/H2 blockers, corticosteroids, albuterol if bronchospasm

What class of dr ith shock due to pulmonary embolism?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to LT, what is causing his hemodynamic condition? What medications should be administered next?

LT has pneumonia and sepsis. Byproducts of sepsis lead to arterial vasodilation, low SVR, and low blood pressure. In addition to giving LT broad-spectrum antibiotics to treat his pneumonia, you give him vasopressors to increase his SVR and blood pressure. After LT is admitted to the ICU, he improves over the course of 3 days. After those 3 days, he is transferred to a general medical unit and then discharged to home.

Summary

- Shock is a state of systemic hypoperfusion resulting in tissue hypoxia, potentially leading to multiorgan failure and death if untreated.
- There are five types of shock: distributive, cardiogenic, hypovolemic, neurogenic, and obstructive, each with distinct causes and hemodynamic profiles.
- Initial treatment avenous fluids to stabilize blood flow and organ perfusion, except in cases of pulmonary edema.
- Distributive shock is managed with vasopressors like norepinephrine, which cause vasoconstriction by stimulating adrenergic receptors.
- Cardiogenic shock is treated with inotropes such as dobutamine and milrinone, which increase cardiac contractility and may cause vasodilation.
- Obstructive shock is treated by addressing the underlying cause, such as thrombolytics for pulmonary embolism or surgical interventions for cardiac tamponade and tension pneumothorax.

Review Questions

1. A 65-year-old female presents with a 2-week history of cough, fever, and chills. She is confused and disoriented. Her blood pressure is 70/50 mm Hg, pulse is 125 beats/min, and temperature is 102.5°F. Exam shows right flank tenderness, and her urinalysis shows numerous white blood cells and bacteria. The white blood cell count is elevated. She is admitted to the intensive care unit. Her blood pressure does not change with 1 L of normal saline. Which of the following medications would be the best next step in management?

A. ○ Digoxin

B. ○ Dobutamine

C. ○ Milrinone

D. ● Norepinephrine

Hide Explanation

The correct answer is norepinephrine (D). Norepinephrine is a vasopressor that stimulates adrenergic α and β receptors to cause vasoconstriction; it is the first-line treatment for septic shock, a form of distributive shock in which the SVR is decreased. Digoxin (A), dobutamine (B), and milrinone (C) are inotropes, which work by increasing cardiac contractility. Because the patient's problem is septic shock with low SVR and normal cardiac function, none of these agents would be appropriate.

2. A 55-year-old male with a history of cardiomyopathy (heart failure with an ejection fraction of 10%) is admitted for consideration of heart transplant. He is found to be in cardiogenic shock with low cardiac output and high SVR. He is started on milrinone and listed for transplant. What is the mechanism of action of this medication?

A. ○ α_1-Adrenergic stimulation

B. ○ β_1-Adrenergic stimulation

C. ○ β_2-Adrenergic stimulation

D. ● PDE3 inhibition

Hide Explanation

The correct answer is PDE3 inhibition (D). Milrinone is an inotrope that is used for cardiogenic shock. It works by inhibiting PDE3, leading to increases cAMP and intracellular Ca^{2+}; the increases in these substances improve cardiac contractility. Milrinone also causes vasodilation. Phenylephrine, epinephrine, and norepinephrine are vasopressors that all induce α_1-adrenergic stimulation (A). Dobutamine causes β_1-adrenergic stimulation (B) to increase heart rate and contractility. Epinephrine, norepinephrine, dobutamine, and dopamine all cause differing degrees of β_2-adrenergic stimulation (C).

3. A 35-year-old female who is 16 weeks pregnant is admitted with confusion and cold extremities. Her blood pressure is 85/75 mm Hg, pulse 110 beats/min, and temperature is 100.4°F. The heart, lung, and abdominal/pelvic exams are normal. Which of the following would be the most appropriate initial treatment?

A. ● Give normal saline

B. ○ Start epinephrine

C. ○ Start milrinone

D. ○ Start norepinephrine

E. ○ Start vasopressin

Hide Explanation

The correct answer is to give normal saline (A). This pregnant patient has shock (hypotension and confusion) with a mild fever and no immediate obvious cause of hypotension. The differential diagnosis is wide, with either hypovolemic or distributive shock most likely. Even before sending her for tests, she should be given 1 to 2 L of normal saline to improve her blood pressure. This should only be withheld in patients with pulmonary edema, which is not present here. If there is no response, pressor agents can then be used. Although epinephrine (B), norepinephrine (D), and vasopressin (E) are useful for distributive shock, this diagnosis is not proven in this patient’s case, and these drugs can cause pregnancy termination. She has no evidence of heart failure, so the inotrope milrinone (C) is not indicated.

4. Which type of shock is caused by sympathetic output?

A. ○ Distributive shock

B. ○ Cardiogenic shock

C. ○ Hypovolemic shock

D. ● Neurogenic shock

E. ○ Obstructive shock

Hide Explanation

The correct answer is pathetic output. Distributive shock (A) arises from systemic vasodilation, often due to sepsis or anaphylaxis. Cardiogenic shock (B) results from heart conditions leading to low cardiac output. Hypovolemic shock (C) is due to low intravascular volume. . Obstructive shock (E) involves blood flow obstruction.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the hemodynamic findings for each type of shock.~~

02 ☑ ~~Explain the use of fluid resuscitation in shock.~~

03 ☑ ~~Justify the use of pharmacological agents in shock treatment~~

04 ☑ ~~Outline the treatment plan for each type of shock~~

Objective Confidence: 4/4